

Question	Marking details	Marks Available					
		AO1	AO2	AO3	Total	Maths	Prac
6	<u>Mutation</u> A1 Mutations are {spontaneous / random events / very low in frequency} A2 The change in structure of {gene / chromosome} or change in {sequence of bases / DNA} A3 Are the source of genetic variation A4 ionising radiations / e.g. gamma radiation / UV / X-rays A5 cigarette smoke / smoking / cigarettes / polycyclic hydrocarbons A6 carcinogens / mutagens / chemicals causing cancer / benzenes / asbestos <u>Gene mutation</u> B1 Ref to point mutation / Ref to {deletion / addition / substitution / inversion} of bases / reference to frame shifts / {nonsense / stop / missense} mutations B2 (Gene mutations) can affect protein synthesis / or description of B3 changing the sequence of amino acid / code for a different amino acid B4 Can change the phenotype of an organism B5 Some mutations have no effect on the phenotype / or description of degenerate code / mutation in introns / silent mutations B6 Example of gene mutation: Cystic fibrosis / sickle cell anaemia <u>Chromosome mutation</u> C1 Chromosomal mutations affect many genes / different sequence of genes (on chromosomes) C2 (Different types of chromosome mutations) e.g. {inversion / translocation / deletion} of part of a chromosome / non-disjunction (in meiosis) C3 Most chromosome mutations occur during crossing over (in prophase-I) C4 Could result in {trisomy / aneuploidy / change in chromosome number} C5 may be fatal C6 Example of chromosome mutation: Downs Syndrome	6	3		9		

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	7-9 marks Indicative content of this level is detailed explanation from all three areas of indicative content. <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Indicative content of this level is detailed explanation from two areas of indicative content or less detail from three areas of indicative content. <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i> 1-3 marks Indicative content of this level is detail from one area of indicative content. <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
	Question 6 total	6	3	0	9	0	0